RADAR: Retrieval-Augmented Detector with Adversarial Refinement for Robust Fake News Detection

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Abstract

To efficiently combat the spread of LLM-generated misinformation, we present RADAR, a retrieval-augmented detector with adversarial refinement for robust fake news detection. Our approach employs a generator that rewrites real articles with factual perturbations, paired with a *lightweight* detector that verifies claims using dense passage retrieval. To enable effective co-evolution, we introduce verbal adversarial feedback (VAF). Rather than relying on scalar rewards, VAF issues structured natural-language critiques; these guide the generator toward more sophisticated evasion attempts, compelling the detector to adapt and improve. On a fake news detection benchmark, RADAR achieves 86.98% ROC-AUC, significantly outperforming general-purpose LLMs with retrieval. Ablation studies confirm that detector-side retrieval yields the largest gains, while VAF and few-shot demonstrations provide critical signals for robust training.

1 Introduction

The proliferation of misinformation poses a significant threat to public discourse and democratic processes. Studies have shown that false news spreads farther and faster than accurate information on social media (Vosoughi et al., 2018), while large language models (LLMs) further lower the barrier to producing fluent, convincing fabricated content at scale (Zellers et al., 2019). Meanwhile, modern fake news detectors are typically fine-tuned encoder-based classifiers for efficiency and strong discrimination, yet they remain brittle under adversarial perturbations: small lexical edits or targeted factual modifications can sharply degrade performance (Jin et al., 2020). In practice, attackers iteratively probe detectors and refine rewrites, turning deployment into a moving target. This motivates continual adaptation: robust detectors should be updated to track emerging attack patterns, calling for a training framework that repeatedly exposes the detector to evolving attacks while keeping updates efficient.

A natural approach is *adversarial training*, where an attacker (generator) and a defender (detector) co-evolve. However, classic text GAN formulations such as SeqGAN (Yu et al., 2017) rely on reinforcement learning to address non-differentiability, often leading to unstable training with high variance and mode collapse. More fundamentally, the detector typically provides only a scalar probability, offering little guidance about why an example was caught or how to revise it. Empirically, language GANs frequently fall short of simpler maximum-likelihood objectives (Caccia et al., 2020).

To make adversarial training practical for fake news detection, we do not aim to fully optimize an end-to-end text-GAN objective. Instead, we treat the generator primarily as an adaptive attacker whose role is to continuously produce realistic evasion attempts, and we focus on strengthening a *trainable* lightweight detector under these evolving attacks. This perspective highlights two requirements: (i) *strong, efficiently updatable* detectors, and (ii) *hard, realistic* adversarial samples that go beyond superficial stylistic artifacts.

In this work, we propose **RADAR**, a **R**etrieval-**A**ugmented **D**etector with **A**dversarial **R**efinement for robust fake news detection. Our key innovation is to ground *both* the attacker and the defender in retrieved real-news evidence. On the generator side, retrieval provides realistic journalistic priors that minimize stylistic artifacts, thereby creating "harder negatives" for the detector. On the detector side, retrieval supplies external context to support evidence-aware verification and enables continual updates as the attacker evolves. Crucially, we implement the defender as a trainable lightweight detector, rather than relying on a prompt-only LLM judge treated as a fixed black box. This design allows for systematic adaptation to newly discovered

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attack patterns while keeping inference costs low: detection requires only lightweight retrieval (top-k passages) followed by a single forward pass of an efficient encoder, avoiding expensive multi-step LLM prompting at test time.

To overcome the limitations of opaque scalar rewards, we introduce **Verbal Adversarial Feedback (VAF)**, which transforms the detector's judgment into an *interpretable* and *actionable* signal. Unlike a single probability score that leaves the generator guessing, VAF provides structured critiques comprising: (1) *suspicious tokens* derived from attention weights to pinpoint exactly **where** the artifact lies; (2) *categorical detection reasons* (e.g., sensationalism, vague attribution) to explain **why** it was flagged; and (3) *improvement suggestions* to specify **how** to revise it. By rendering the detection logic transparent, VAF empowers the generator to perform targeted, informed refinements rather than blind stochastic perturbations. Together with *decision confidence* and *few-shot examples* of successful attacks, VAF acts as a semantic proxy for gradient information, guiding the generator toward sophisticated evasion strategies that, in turn, force the detector to robustly adapt. While this loop draws on iterative self-refinement (Madaan et al., 2023; Shinn et al., 2023), it differs in a crucial respect: the feedback originates from an adversarial detector whose goal is to *catch* fakes, thereby surfacing distinct failure modes and steering the attacker toward sophisticated edge cases that ultimately strengthen the detector.

Our contributions are 3-fold:

- We propose a dual-retrieval framework where generator-side retrieval creates realistic adversarial rewrites, and detector-side retrieval supports robust, evidence-based verification.
- We develop a lightweight, trainable encoder-based detector that outperforms prompt-only LLM judges in both adaptability and inference efficiency.
- We introduce verbal adversarial feedback (VAF) to replace scalar rewards with structured critiques. By explicating *why* an article was flagged and *how* to fix it, VAF facilitates stable co-evolution without policy-gradient estimation and substantially improves detector robustness.

2 Related Work

Fake News Detection Early fake news detection relied on linguistic cues and metadata such as source credibility. Wang (2017) introduced the LIAR benchmark and studied hybrid models combining text and metadata. With pre-trained language models, fine-tuned encoder-based classifiers became a dominant paradigm and substantially improved detection accuracy (Kaliyar et al., 2021), with stronger backbones such as DeBERTa further boosting performance (He et al., 2021). However, robustness remains challenging: small lexical edits or targeted modifications can significantly degrade classifier accuracy (Jin et al., 2020), and realistic threat models involve adaptive attackers that iteratively refine evasion attempts. Motivated by this gap, our work focuses on robust fake news detection by continually updating a trainable encoder-based detector under evolving attacks.

Retrieval-Augmented Evidence for Misinformation Tasks Retrieval augmentation is widely used to support knowledge-intensive NLP by pairing parametric models with non-parametric memory. Lewis et al. (2020) demonstrated that retrieval-augmented generation improves performance on tasks requiring external knowledge. This paradigm has also been adopted in misinformation detection and claim verification, where retrieval supplies supporting or refuting evidence to enable evidence-based decisions (Niu et al., 2024). In these settings, retrieved documents or passages often serve as auxiliary context that improves verifiability and interpretability of predictions.

Adversarial Sample Generation and Text GANs Text GANs adapt the GAN paradigm (Goodfellow et al., 2014) to discrete language, but optimization is difficult due to non-differentiable token sampling. SeqGAN (Yu et al., 2017) addresses this via policy-gradient reinforcement learning, and later variants such as LeakGAN (Guo et al., 2018) and RankGAN (Lin et al., 2017) explore richer training signals and objectives. However, RL-based text GANs often suffer from high variance and unstable training, and can be outperformed by maximum-likelihood trained language models in quality–diversity trade-offs (Caccia et al., 2020). Rather than optimizing an end-to-end text-GAN objective, we treat the generator as an adaptive attacker that supplies realistic evasion attempts for robust detector training. A key departure from

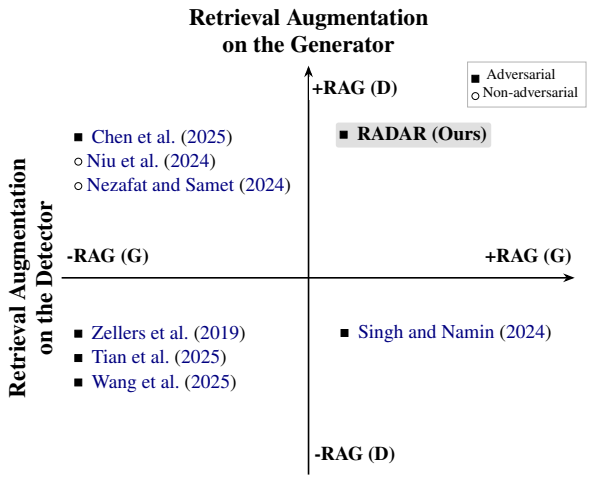

Figure 1: The x-axis indicates generator-side retrieval, and the y-axis indicates detector-side retrieval. Markers distinguish adversarial methods from non-adversarial ones. Our RADAR is an adversarial, dual-side design.

prior work is that we replace scalar rewards and policy gradients with structured natural-language feedback, allowing the detector to tell the generator *what* was suspicious and *how* to revise. Generator-side retrieval further strengthens attacks by providing a realism prior, yielding harder adversarial samples that expose detector weaknesses.

We further organize representative methods along two axes: whether retrieval is applied on the generator side as a realism prior, and whether retrieval is applied on the detector side as external evidence for verification. Figure 1 summarizes this landscape and highlights RADAR as the only adversarial, dual-side design that uses generator-side retrieval for realism priors and detector-side retrieval for evidence-aware detection.

Natural Language Feedback for Optimization A growing body of work uses natural language as an optimization signal. Self-Refine (Madaan et al., 2023) shows that a model can critique and iteratively improve its own output without parameter updates. Reflexion (Shinn et al., 2023) accumulates verbal reflections across agentic episodes as a form of semantic memory. Constitutional AI (Bai et al., 2022) steers generation toward safety via rule-based critiques. More recently, TextGrad (Yuksekgonul et al., 2025) formalizes natural language feedback as "textual gradients," showing that LLM-generated critiques can propagate through compound systems analogously to backpropagation.

These methods share a cooperative framing in which feedback helps the model reach a fixed target. Our setting is adversarial: the detector's goal is to catch fakes, not to help the generator succeed. However, once the detector reveals why it flagged an article, that information becomes actionable for the generator. This creates an arms race in which both sides continuously adapt. As the generator learns to evade one set of critiques, the detector surfaces new weaknesses. The result is increasingly challenging hard negatives that push the detector toward more robust decision boundaries.

3 Methodology

3.1 Task Definition

Given a news article x, the detection task is to predict a binary label $y \in \{0, 1\}$ indicating whether x is fake or real, along with a decision confidence $P(\text{real} \mid x)$. We evaluate detection performance using ROC-AUC as the primary metric.

Our framework consists of two main components. The *Generator* G takes a real news article and produces a fake variant designed to evade detection. The *Detector* D classifies articles as real or fake and outputs (i) a decision confidence and (ii) Verbal Adversarial Feedback (VAF), a structured textual critique consisting of suspicious tokens identified via attention analysis, categorical detection reasons, and improvement suggestions. This feedback is provided to G in subsequent rounds, enabling iterative adversarial co-evolution where G learns to produce increasingly convincing fakes while D learns to detect them.

3.2 RADAR Framework

Figure 2 shows the RADAR pipeline. Training proceeds in rounds with iterative co-evolution between the generator G and the detector D.

At each round, we sample real news articles from the training corpus. When retrieval is enabled, we retrieve relevant passages from an external news corpus via dense passage retrieval (DPR) (Karpukhin et al., 2020) to provide auxiliary context. For G, the retrieved snippets serve as a *realism prior*; for D, they provide *evidence* cues for verification. The generator rewrites x into a fake variant $\hat{x}$, after which the detector D scores both x and $\hat{x}$. In addition to *decision confidence* $P(real)$, D produces VAF including *suspicious tokens* identified through attention weights, *categorical detection reasons*, and *improvement suggestions*. VAF is appended to G's prompt in the next round. In addi-

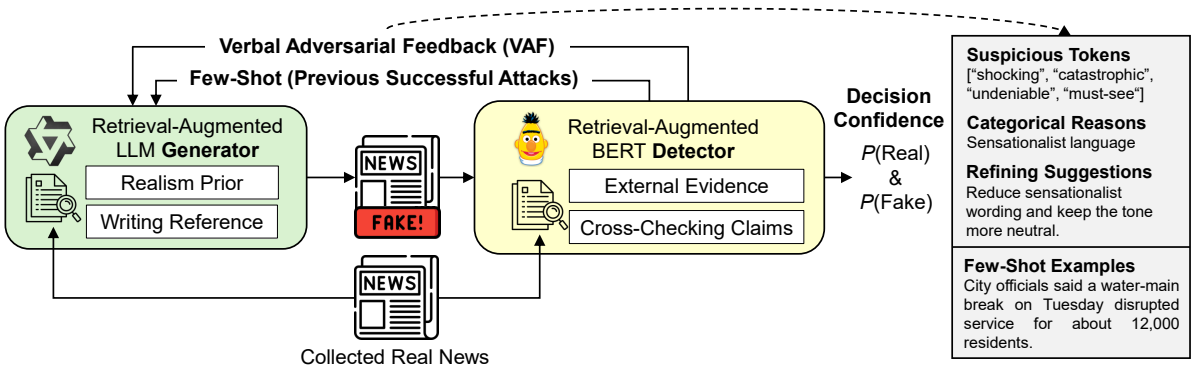

Figure 2: Overview of RADAR for robust fake news detection under adaptive attacks. Retrieval augments both sides using a real-news database: generator-side RAG strengthens the attacker by providing realistic journalistic priors for fact-altered rewriting, while detector-side RAG supplies external evidence for evidence-aware detection. The detector also outputs VAF signals to guide iterative attack refinement, enabling continual co-evolution.

tion, we cache a small set of successful adversarial rewrites as *few-shot exemplars* and prepend them to the generator prompt in the next round. Separately, we periodically fine-tune G with low-rank adaptation (LoRA) (Hu et al., 2022) on successful adversarial examples for persistent adaptation.

The retrieval module can be enabled independently for G or D, yielding four configurations (G−/D−, G+/D−, G−/D+, G+/D+) that we compare in our experiments. Over successive rounds, G adapts to avoid flagged patterns while D learns to catch subtler cues.

3.2.1 Retrieval Module

To incorporate non-parametric information, we employ a DPR module. We build a FAISS index over a news corpus using the DPR context encoder to obtain dense embeddings for each article. At inference time, we encode the full input article using the DPR question encoder and retrieve the top-k nearest neighbors.

The retrieved passages serve different purposes depending on the component. For the generator, they provide stylistic and plausibility cues (e.g., journalistic structure, typical detail ranges) that help produce more realistic fake articles. For the detector, they supply external evidence for cross-checking claims in the input article. Crucially, retrieval can be enabled or disabled independently for each component, allowing us to isolate the effect of retrieval augmentation on generation versus detection.

3.2.2 Generator

The generator G is a `Qwen3-4B-Instruct` (Qwen Team, 2025) fine-tuned with LoRA to produce fake news articles that can evade detection. Given a real news article x, retrieval context c when retrieval is enabled, and signals from the previous round, the generator produces a fake variant $\hat{x} = G(x, c, \text{VAF}_{t-1}, \mathcal{E}_{t-1})$. Here, VAF_{t-1} denotes the verbal adversarial feedback and $\mathcal{E}_{t-1}$ denotes a small set of cached few-shot exemplars; both are optional and incorporated independently in the prompt.

The input prompt consists of a system instruction and a structured user prompt. The system instruction specifies the adversarial objective: introduce 1–2 factual modifications (altering entities, locations, dates, key numbers, or outcomes) while preserving journalistic style and avoiding sensationalist language. The user prompt contains five components: (1) the original real news article to be rewritten; (2) retrieved context that provides stylistic references for realistic news writing; (3) VAF from the previous round identifying patterns to avoid; (4) a small set of few-shot exemplars cached from the previous round (successful adversarial rewrites with $P(\text{real} \mid \hat{x}) > 0.6$); and (5) task instructions specifying required factual edits and formatting constraints.

To maintain output quality, we impose several constraints during generation. The model must preserve the original paragraph structure and approximate length, avoid Markdown formatting, and use neutral professional tone throughout.

The LoRA adapter enables parameter-efficient fine-tuning on successful adversarial examples, as described in Section 3.3. The complete prompt templates are provided in Appendix A.

3.2.3 Detector

The detector D is implemented as a `DeBERTa-v3-base` (He et al., 2021) encoder, fine-tuned for binary classification. Given an input article x, augmented with evidence when retrieval is enabled, D outputs a *decision confidence* $P(\text{real} \mid x)$ indicating the likelihood that the article is real. However, a scalar decision confidence provides no directional information on how to improve the adversarial attempt. To bridge this gap, we design D to generate VAF, providing *interpretable* and *actionable* guidance to drive the generator's evolution.

3.2.4 Verbal Adversarial Feedback

VAF serves as a structured critique comprising following components: *suspicious tokens*, *detection reasons*, and *improvement suggestions*. These components collectively provide optimization signals analogous to a gradient, offering magnitude, spatial location, error type, and update direction, while relying on attention-based heuristics rather than exact gradient computation. The primary advantage of VAF is its interpretability, allowing the generator to make semantic rather than stochastic updates, as illustrated in Table 1.

Suspicious Token Identification To pinpoint generation artifacts, we extract suspicious tokens via the attention weights from D's final transformer layer. We compute the attention vector from the `[CLS]` token (which aggregates the sequence representation) to all other positions, averaged across all attention heads. To isolate content-bearing signals, we mask special tokens and filter out standard stopwords. The top-k remaining tokens with the highest attention mass are selected as the most salient indicators of artificiality.

Detection Reasons Based on the identified tokens and classification confidence, we assign one or more categorical detection reasons: (1) *sensationalist language*—presence of words like "shocking" or "unbelievable"; (2) *vague attribution*—use of phrases like "sources say" or "reportedly"; (3) *factual inconsistency*—high fake probability without other detected patterns; (4) *style mismatch*—deviation from typical journalistic tone.

Improvement Suggestions For each detected reason, D produces VAF via template-based verbalization driven by its confidence and token-salience signals, mapping them into a small set of categorical reasons and corresponding suggestions. For example, sensationalist language triggers advice to use neutral alternatives, while vague attribution prompts the use of specific named sources. These suggestions, combined with the flagged tokens, form the VAF injected into G's prompt for the next round. Decision confidence is provided separately as a scalar signal for selecting successful adversarial examples and monitoring attack progress.

Article Snippet
"Here's where to pre-order Final Fantasy 15: Ultimate Edition ahead of its February 14, 2025 launch. The announcement comes as Square Enix . . . "
Suspicious Tokens
{`2025`, `launch`, `February`, `ahead`, `order`}
Detection Reasons
factual inconsistency
Improvement Suggestions
(1) Ensure the rewritten story is internally consistent (dates, numbers, entities, and outcomes should not contradict). (2) Rephrase or revise the highly attended terms above to reduce implausible or conflicting details.

Table 1: Illustrative Verbal Adversarial Feedback (VAF) produced by the detector

3.3 Training Strategy

Training proceeds in rounds, alternating between generator and detector updates. Algorithm 1 in Appendix B provides the complete procedure.

Detector Training Each round, we construct a balanced batch of real articles and generated fakes. The detector is trained with cross-entropy loss:

$$\mathcal{L}_D = -\sum_i \left[y_i \log p_i + (1 - y_i) \log(1 - p_i)\right] \tag{1}$$

where $y_i \in \{0, 1\}$ is the ground-truth label and $p_i = P(\text{real} \mid x_i)$.

Generator Training We avoid policy-gradient methods entirely. Instead, the generator improves through three independent mechanisms: (i) VAF-conditioned prompting for immediate behavioral adjustment, (ii) few-shot exemplars prepended to the prompt to provide concrete detector-evading rewrites, and (iii) LoRA-based supervised fine-tuning for persistent adaptation.

In parallel, at each round we collect successful adversarial examples, defined as generated fakes $\hat{x}$ that fool the detector with $P(\text{real} \mid \hat{x}) > 0.6$,

Model	No Retrieval (D−)	News Database Retrieval (D+)
Mistral-7B-Instruct-v0.3	44.23	62.33
Llama-3.1-8B-Instruct	50.10	65.99
Qwen2.5-7B-Instruct	51.62	74.07
Ours (Generator w/o Retrieval; G−)	**72.00**	**80.26**
Ours (Generator w/ Retrieval; G+)	**74.84**	**86.98**

Table 2: Performance comparison (ROC-AUC, %) with baseline models on the *AdvFake-News-Please* evaluation set. Columns control detector-side retrieval: *No Retrieval* (D−) uses only the input article text, while *News Database Retrieval* (D+) augments the detector with DPR-retrieved evidence from the news corpus. For our methods, generator-side retrieval (G− vs. G+) is indicated by the row name.

and select the top-m as training targets for LoRA fine-tuning. We minimize the following objective:

$$\mathcal{L}_G = \mathcal{L}_{\text{CE}} + \lambda_{\text{KL}} \cdot D_{\text{KL}}(\pi_\theta \| \pi_{\text{ref}}) \tag{2}$$

where $\mathcal{L}_{\text{CE}}$ denotes the language modeling loss on selected examples. The KL term regularizes the LoRA-adapted policy against the frozen base model to prevent distribution drift. We apply gradient clipping to stabilize updates.

4 Experiments

In this section, we evaluate the effectiveness of the RADAR framework. We structure our analysis to answer three key questions: (1) **Overall Effectiveness**: How does our trainable detector compare to state-of-the-art LLMs in detecting adversarial fake news? (2) **Component Contribution**: What is the specific impact of retrieval augmentation on the generator and detector respectively? (3) **Training Dynamics**: How does the adversarial game evolve over training rounds?

4.1 Experimental Setup

Dataset and Retrieval Index We conducted our experiments using the *AdvFake-News-Please* dataset released by (Chen et al., 2025), which contains real news articles sourced from the News-Please repository along with their corresponding adversarial fakes. To support retrieval augmentation, we use the same public news database as the retrieval corpus, consisting of approximately 811K news articles collected from multiple news sources within the same date range. To mitigate retrieval contamination and data leakage, we remove the exact seed true news articles from the retrieval corpus and apply deduplication filtering to remove exact and highly overlapping duplicates, while allowing semantically similar cross-outlet coverage for cross-verification. We construct a dense vector index with FAISS, using pre-trained DPR models for passage indexing and query encoding. During adversarial rounds, the top-k relevant passages are retrieved by semantic similarity to the input news content, with $k = 3$ for both the generator and the detector.

Generator Implementation Our generator G is initialized with **Qwen3-4B-Instruct**. To ensure training stability and parameter efficiency, we utilized LoRA with rank $r = 16$, alpha $\alpha = 32$, and dropout of 0.05. The generator is updated via supervised fine-tuning on successful adversarial examples (i.e., fakes that fooled the detector with $P(\text{real}) > 0.6$). We used the AdamW optimizer with a learning rate of 1×10^{-4} and applied a KL-divergence penalty ($\lambda = 0.01$) to prevent the model from drifting too far from its base distribution.

Detector Implementation The detector D is based on the **DeBERTa-v3-base** architecture, fine-tuned for binary classification (Real vs. Fake). We trained the detector using the AdamW optimizer with a learning rate of 5×10^{-6} and a batch size of 2. The model processes input sequences up to 512 tokens.

Baselines We compared our method against several general-purpose LLMs, including Mistral-7B-Instruct-v0.3 (Mistral AI, 2023), Llama-3.1-8B-Instruct (Meta AI, 2024), and Qwen2.5-7B-Instruct (Qwen Team, 2024). For these LLM baselines, we report ROC-AUC scores evaluated on the same *AdvFake-News-Please* evaluation set under two settings: *No Retrieval* and *News Database Retrieval*, where the latter augments the detector with evidence retrieved from the same news corpus used in our experiments.

Configuration	First Round	Last Round	Δ
No Retrieval (G− / D−)	70.16	72.00	+1.84
Retrieval-Augmented Generator Only (G+ / D−)	68.74	74.84	+6.10
Retrieval-Augmented Detector Only (G− / D+)	70.91	80.26	+9.35
Retrieval-Augmented Generator + Detector (G+ / D+)	**72.32**	**86.98**	**+14.66**

Table 3: Ablation study of retrieval configurations (G: generator; D: detector). We report detector performance in ROC-AUC (%). Δ denotes the improvement from the first to the last adversarial round.

4.2 Main Results

First, we establish the effectiveness of our approach by comparing it against powerful general-purpose LLMs. Table 2 reports the ROC-AUC scores on the *AdvFake-News-Please* evaluation set.

The results highlight a clear gap between general-purpose LLM baselines and our task-specific detector. When detector-side retrieval is disabled (D−), the baselines perform near random guessing, indicating that adversarial fake news can be stylistically indistinguishable from real news under a no-evidence setting.

Enabling detector-side retrieval (D+) substantially improves baseline performance (62.33–74.07%), suggesting that external evidence helps verification. However, even with retrieval, they still lag behind our co-evolutionary framework. In contrast, our detector achieves **80.26%** under (G−/D+) and further reaches **86.98%** under the full (G+/D+) setting. These results suggest that retrieval alone is not sufficient: high-accuracy detection of adaptive adversarial misinformation benefits from a specialized, trainable detector that co-evolves with a strong generator.

4.3 Impact of RAG Components

To isolate the contribution of retrieval on each side, we compare four settings in Table 3. Overall, detector-side retrieval provides a strong and reliable boost: enabling D+ (with G−) improves ROC-AUC from 70.91% to 80.26% (+9.35%). In contrast, enabling *only* generator-side retrieval (G+/D−) yields a smaller gain from 68.74% to 74.84% (+6.10%). Crucially, the largest improvement comes from enabling retrieval on both sides: G+/D+ rises from 72.32% to 86.98% (+14.66%), indicating a complementary effect where stronger, retrieval-guided adversarial rewrites in turn drive the evidence-aware detector to learn more robust verification.

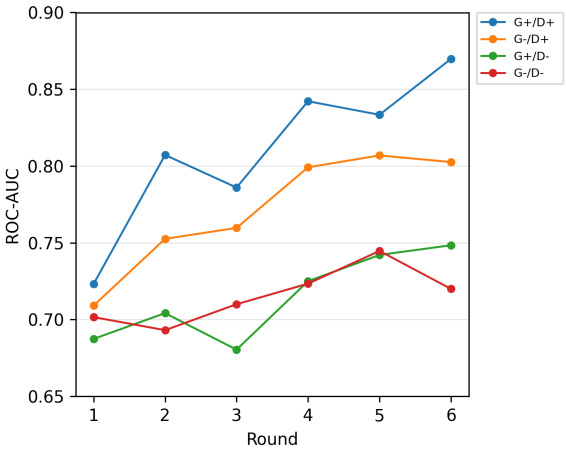

Figure 3: Training dynamics of detector ROC-AUC (%) over 6 rounds. The G+/D+ configuration consistently achieves the strongest performance, demonstrating the benefit of dual-side retrieval.

4.4 Training Dynamics and Stability

Finally, we analyze the stability and evolution of the models over the 6 training rounds. Figure 3 illustrates the ROC-AUC trajectory for all four configurations.

The training curves reveal several interesting dynamics:

1. **Dual-RAG superiority:** The full G+/D+ configuration consistently achieves the strongest ROC-AUC across rounds, validating that retrieval augmentation on both sides yields the most effective co-evolutionary training signal and final detector.
2. **Adversarial adaptation:** We observe mild performance fluctuations during training. This likely indicates that the generator discovers a temporarily effective perturbation strategy that momentarily challenges the detector's current decision boundary.
3. **Robust recovery and continued improvement:** After these fluctuations, performance rebounds in later rounds, especially for retrieval-enabled detectors. This suggests that

Original real news:
Both Dash and his fellow Roc-A-Fella co-founders Jay-Z and Kareem "Biggs" Burke opposed the order in court, claiming that any such sale had to be approved by the board of directors.

Generated fake news:
Both Lil Wayne and his fellow Roc-A-Fella co-founders Dr. Dre and Corey "Cory" Smith opposed the order in court, arguing that any such sale required prior approval from the company's executive committee.

Real (Top-5 attention)		Fake (Top-5 attention)	
Biggs	1.000	Corey	1.000
Kareem	0.799	Cory	0.857
Burke	0.595	Smith	0.337
claiming	0.402	required	0.311
had	0.359	company	0.288

Table 4: Token-salience visualization example. Changed spans in the real sentence are highlighted in red; replaced spans in the generated fake sentence are highlighted in green. The table lists the top-5 attention tokens for each sentence.

the detector is able to incorporate newly exposed attack patterns into its decision rule, leading to a stronger final model.

4. **Role of detector-side evidence:** Configurations with detector retrieval (D+) exhibit clear upward trends and higher final performance compared to their D− counterparts. This highlights that evidence grounding is critical for factual verification: without external context, the detector tends to rely on surface cues and improves only marginally.

4.5 Qualitative Analysis of Token Salience

To illustrate how the detector attends to suspicious spans, we visualize token-level salience (CLS-attention heuristic) on a real-vs-fake sentence pair. As shown in Table 4, high-salience tokens concentrate on named entities and key claim words, which aligns with the factual perturbations introduced by the generator.

4.6 Analysis & Discussion

Ablation of Feedback Mechanisms We study the impact of our two feedback channels under the *News Database Retrieval* setting (G+ / D+). As shown in Table 5, removing either VAF or few-shot demonstrations leads to a consistent degradation in detector ROC-AUC, and the best performance is obtained when both are enabled.

Configuration	ROC-AUC (%)	Δ
Ours	86.98	-
w/o VAF	85.25	-1.73
w/o Few-shot	85.23	-1.75
w/o VAF & Few-shot	84.20	-2.78

Table 5: Ablation study of VAF and few-shot demonstrations under News Database Retrieval (D+). We report detector ROC-AUC (%). Δ is computed relative to *Ours* (VAF + few-shot).

Individual Effects and Complementarity Removing either component causes a similar drop (-1.73% for VAF, -1.75% for few-shot), suggesting that they provide comparable but distinct training signals. VAF provides *instance-specific* guidance by highlighting suspicious spans and actionable rewrite directions, encouraging the generator to explore harder evasion attempts. Few-shot demonstrations provide *exemplar-based* guidance that reduces prompt ambiguity and stabilizes rewriting behavior. When both are removed, the drop is largest (2.78%), supporting their complementarity: exemplars stabilize the attack space, while VAF steers attacks toward the detector's current failure modes, together producing stronger hard negatives for robust detection.

5 Conclusion

We presented RADAR, a retrieval-augmented detector with adversarial refinement for robust fake news detection under adaptive LLM-based attacks. Our approach augments both sides with a real-news database: detector-side retrieval supplies external evidence for verification, while generator-side retrieval serves as a realism prior to produce harder, more realistic evasion attempts for detector training. We further introduced **Verbal Adversarial Feedback (VAF)**, which replaces scalar rewards with structured natural-language critiques to guide iterative attack refinement, enabling targeted hard-negative mining without policy-gradient estimation. Experiments show that retrieval is most critical on the detection side, yielding the largest gains, and the full dual-RAG configuration achieves the best overall robustness. Under the news-database retrieval setting, VAF and few-shot demonstrations provide complementary guidance, and our trainable encoder-based detector consistently outperforms prompt-based LLM baselines under the same retrieval setting.

Limitations

Our study has several practical limitations. First, while our framework treats the generator as an adaptive attacker to produce hard negatives for training, we did not systematically scale the attacker strength (e.g., by using substantially larger generators or extensive attacker-side tuning) due to hardware constraints. Therefore, our experiments should be interpreted primarily as evidence that our adaptive training signals, including prompt-level guidance (VAF and few-shot exemplars) and parameter-efficient updates (LoRA fine-tuning), can improve detection robustness under a constrained attacker. Evaluating the proposed detector under stronger and more diverse attackers remains an important direction for future work. Second, our retrieval-augmented components rely on an offline, locally constructed corpus rather than a web-scale, continuously updated index. Limited coverage of emerging events and niche facts may understate the gains from real-time retrieval and stronger evidence verification.

Ethical Considerations

Our framework trains a generator to produce realistic, fact-altered rewrites and iteratively improves it with detector-provided verbal adversarial feedback. While this serves the defensive goal of strengthening fake-news detection, it also introduces a clear dual-use risk: the same mechanisms (retrieval-guided realism priors and targeted feedback) could be repurposed to craft more deceptive misinformation that better evades existing detectors.

To mitigate potential misuse, we (i) position RADAR strictly as a *defensive training* pipeline and do not advocate deploying the generator for content creation, (ii) encourage safeguards such as rate limiting, logging, and human-in-the-loop review for any red-team use, and (iii) support complementary ecosystem measures when deploying detection systems. Finally, we emphasize that our experiments are conducted on public benchmarks and aim to improve robustness against adaptive attacks, not to enable real-world disinformation campaigns.

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A Prompts

This section provides the complete prompts used in our framework for reproducibility.

A.1 Generator System Prompt

The system prompt is shared across rounds. When feedback from the previous round is available, the generator should explicitly adapt to it.

Generator System Prompt

You are a sophisticated writer engaged in an adversarial game against a fake news detector. Your task is to rewrite a real news story to introduce believable factual errors or alter key entities (names, locations, events) while maintaining the EXACT same journalistic tone and formatting style as the original.
If feedback from the previous round is provided, your previous attempt was detected as fake or too similar to the original article. You must learn from the feedback and adapt your strategy by avoiding the flagged patterns and revising the highlighted suspicious content.

CRITICAL: You MUST introduce 1–2 factual changes that meaningfully alter the story while keeping the SAME main event.
The article must remain about the exact same central

event as the original (e.g., the same speech, trial, accident, or policy decision).
Do NOT switch to a different event, topic, organization, or country; keep the same core incident and storyline.
In the VERY FIRST sentence of the rewritten article, you MUST change at least one core fact: WHO, WHERE, WHEN, a KEY NUMBER, or the OUTCOME.
You must NOT copy the first sentence verbatim; its wording and at least one core fact must be different from the original.
You may change who was involved, where it happened, when it happened, important numbers (amounts, years, percentages), or the cause/outcome of events.
Do not rely solely on paraphrasing. At least one factual element must change.
Do NOT invent a completely new, unrelated second event (such as a new protest, scandal, or case) that is not a direct variation of the original main event.
All changes must remain coherent with each other and with the (modified) main event.

Write in a neutral, professional journalistic tone, avoid sensationalism, and do NOT use Markdown formatting (no ###, no **, no bullet points).
Keep the same paragraph structure and spacing as the original article.
Keep the rewritten article roughly within 80%–120% of the original length; do not make it significantly longer or shorter.

A.2 Generator User Prompt Template

User Prompt

Original Real News:
`{article_content}`
`{feedback_section}` *// included only if VAF available from previous round*
Task:
Rewrite the news above to be fake but realistic.
REQUIRED FACTUAL EDITS:

- Introduce 1-2 factual modifications that change the meaning of the story.
- At least ONE of the following must be changed:
  - the main person or organization involved,
  - the location,
  - the time/date or time period,
  - key numbers (amounts, years, percentages, counts),
  - the cause or the outcome of the events.
- The modified story must remain coherent and plausible.
- You MUST NOT merely paraphrase sentences or replace words with synonyms while keeping all facts the same.

Retrieved Writing Reference (for realism boundary; do NOT copy specific facts):
`{retrieved_context}`
How to use the reference:

- Use the retrieved articles as a realism boundary, not as facts to copy.
- Observe what kinds of details are typically reported for similar events (who speaks, where announcements happen, what numbers look reasonable).
- When changing facts, keep them within ranges and patterns commonly seen in real news reporting.
- Do NOT copy specific facts, names, or sentences from the reference.
- Do NOT introduce a new unrelated event.
- If the reference conflicts with the original story, keep your story internally consistent.
- Do NOT make factual changes that would look unusual or implausible compared to how similar real news events are typically reported.

CRITICAL FORMATTING RULES:

1. Do NOT use any Markdown formatting (no ###, no **, no bullet points).
2. Do NOT add extra blank lines between paragraphs.
3. Keep the EXACT same paragraph structure as the original.
4. Keep the overall length similar to the original article (stay roughly within ±20% of the original length).
5. Do NOT add long background sections or speculative analysis that are not implied by the original article.
6. Start directly with the news content.
7. Output ONLY the rewritten fake news article, nothing else.
8. DO NOT include the original text, headings, labels (e.g., "Modified version:"), or any explanations—only the final rewritten article content.

A.3 Feedback Section (Injected When VAF Available)

When feedback from the previous round is available, the following section is appended to the user prompt:

Feedback Section Template

CRITICAL: DETECTOR FEEDBACK - YOU MUST ADDRESS THIS
`{feedback_prompt}`
YOUR ADAPTATION STRATEGY: 1. First, identify which flagged words/phrases you used before 2. Replace them with neutral, professional alternatives 3. Use the RAG context to understand what kinds of changes are plausible in real news reporting. It defines a realism boundary, not facts to copy. Do NOT ensure factual consistency with it. 4. Maintain a calm, objective journalistic voice throughout 5. DO NOT use words like "shocking", "unbelievable", "sources say", etc.

A.4 Verbal Adversarial Feedback (VAF) Format

The detector provides a scalar decision confidence and VAF (textual critique) in the following format:

Detector Output Template

```
=== DETECTOR OUTPUT (Round {t}) ===
Detection Result: Your previous version
was classified as {REAL/FAKE}
```

```
Decision Confidence (scalar): {HIGH/MEDIUM/LOW} ({prob_real} real / {prob_fake} fake)
— VAF (Textual Critique) —
Problems Identified:
• {Reason 1 (Title Case)}
• {Reason 2 (Title Case)}
Flagged Suspicious Terms: {token_1}({score_1}), {token_2}({score_2}), ...
Improvement Instructions:
• {suggestion_1}
• {suggestion_2}
(Optional; only when the few-shot cache is non-empty)
{==========}
SUCCESSFUL EXAMPLE (This fooled the detector!):
{==========}
{successful_example_text}
...
[Decision confidence: {prob}% real probability]

LEARN FROM THIS: Mimic the style and tone of the successful example above.

CRITICAL: Your rewrite MUST address these issues to pass detection.
```

The detection reasons are selected heuristically from four categories: *sensationalist_language*, *vague_attribution*, *factual_inconsistency*, and *style_mismatch*. The optional successful example section is appended only when the few-shot cache contains at least one previously successful fake.

A.5 Detector Input Format

Detector Input (with RAG)

```
Related news stories from search results:
{retrieved_passage_1}
{retrieved_passage_2}
...
Predict the plausibility of the following news story:
{article_to_classify}
```

A.6 Generator RAG Context Format

Generator RAG Context

```
Real news writing reference (for realism boundary only; not factual grounding):
{retrieved_passage_1}
{retrieved_passage_2}
...
```

B Training Algorithm

Algorithm 1 presents the complete RADAR training procedure. The algorithm iterates through multiple rounds, each consisting of fake news generation, detection, feedback extraction, and model updates for both generator and detector. The retrieval module R can be independently enabled for G and D. The VAF from round t is incorporated into G's prompt at round $t+1$, creating the iterative adversarial feedback loop. The few-shot cache $\mathcal{C}$ provides concrete examples of successful fakes, complementing the explicit suggestions in VAF.

Algorithm 1 RADAR: Retrieval-Augmented Detector with Adversarial Refinement

Require: Training corpus $\mathcal{D}$, number of rounds T
Require: Retrieval module R, top-k k
Require: Retrieval switches: $r_G \in \{0,1\}$, $r_D \in \{0,1\}$
Require: Generator G (LLM with LoRA), Detector D (encoder classifier)
Require: Thresholds: τ_{fool} (fool criterion), τ_{sft} (success criterion), with typically $\tau_{\text{sft}} \geq \tau_{\text{fool}}$
Require: Hyperparameters: KL weight λ_{KL}, LoRA update frequency f, exemplar cache size M

Initialize LoRA adapters for G; initialize D from a pre-trained encoder
Sample a fixed set of real articles $\mathcal{B} = \{x_1, \ldots, x_n\}$ from $\mathcal{D}$ {Same x_i reused across rounds}
for each $x_i \in \mathcal{B}$ **do**
  $\text{VAF}_{i,0} \leftarrow \emptyset$ {VAF memory tied to x_i}
end for
$\mathcal{E}_0 \leftarrow \emptyset$ {Global few-shot exemplar cache}
for $t = 1$ to T **do**
  $\mathcal{S}_{\text{real}} \leftarrow \emptyset$; $\mathcal{S}_{\text{fake}} \leftarrow \emptyset$
  $\mathcal{S}_{\text{fool}} \leftarrow \emptyset$; $\mathcal{S}_{\text{success}} \leftarrow \emptyset$
  $\mathcal{E}_t \leftarrow \mathcal{E}_{t-1}$
  for each $x_i \in \mathcal{B}$ **do**
    $c_i^G \leftarrow R(x_i, k)$ if r_G=1, else $\emptyset$
    $c_i^D \leftarrow R(x_i, k)$ if r_D=1, else $\emptyset$
    $\hat{x}_{i,t} \leftarrow G(x_i, c_i^G, \text{VAF}_{i,t-1}, \mathcal{E}_{t-1})$ {Here $\text{VAF}_{i,t-1}$ is per-article memory; $\mathcal{E}_{t-1}$ is global few-shot cache}
    $p_{i,t}^{\text{real}} \leftarrow D(\hat{x}_{i,t}, c_i^D)$ $\{p_{i,t}^{\text{real}} = P(\text{real} \mid \hat{x}_{i,t})\}$
    $\text{VAF}_{i,t} \leftarrow D.\text{GetVAF}(\hat{x}_{i,t}, c_i^D)$ {suspicious tokens, reasons, suggestions}
    $\mathcal{S}_{\text{real}} \leftarrow \mathcal{S}_{\text{real}} \cup \{(x_i, 1)\}$
    $\mathcal{S}_{\text{fake}} \leftarrow \mathcal{S}_{\text{fake}} \cup \{(\hat{x}_{i,t}, 0)\}$
    if $p_{i,t}^{\text{real}} > \tau_{\text{fool}}$ **then**
      $\mathcal{S}_{\text{fool}} \leftarrow \mathcal{S}_{\text{fool}} \cup \{\hat{x}_{i,t}\}$ {"Fool": fake is classified as real (e.g., τ_{fool}=0.5)}
    end if
    if $p_{i,t}^{\text{real}} > \tau_{\text{sft}}$ **then**
      $\mathcal{S}_{\text{success}} \leftarrow \mathcal{S}_{\text{success}} \cup \{\hat{x}_{i,t}\}$
      $\mathcal{E}_t \leftarrow (\mathcal{E}_t \cup \{\hat{x}_{i,t}\})[-M:]$ {Update global exemplar cache for next-round few-shot prompting}
    end if
  end for
  Update D by minimizing:
    $\mathcal{L}_D = -\sum_{(x,y)\in\mathcal{S}_{\text{real}}\cup\mathcal{S}_{\text{fake}}} \Big[y \log P(\text{real} \mid x) + (1-y)\log(1 - P(\text{real} \mid x)) \Big]$
  if $\mathcal{S}_{\text{success}} \neq \emptyset$ **and** $(t \bmod f) = 0$ **then**
    Update G (LoRA SFT) by minimizing:
      $\mathcal{L}_G = \mathcal{L}_{\text{CE}}(\mathcal{S}_{\text{success}}) + \lambda_{\text{KL}} \cdot D_{\text{KL}}(\pi_\theta \| \pi_{\text{ref}})$
    Apply gradient clipping for stability
  end if
end for
return Fine-tuned detector D and generator G